

MediTrace v2026.3: A Longitudinal, Interactive and Visual Medical Record for Patient Empowerment, Pharmacovigilance and Clinical Decision Support

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Abstract

MediTrace is an open-source longitudinal medical tracking tool designed to centralise, visualise and analyse a patient's medical history from heterogeneous PDF documents. Version 2026.3 introduces: client-side OCR via Tesseract.js for scanned documents, a bacteriology module for culture results and antibiograms, radar (spider-web) charts for normalised parameter overview, Progressive Web App (PWA) architecture for offline use and mobile installation, an extensible parameter system with custom source memory, and a streamlined workflow for Claude.ai integration without API key. Applied to a real clinical case covering 2016-2026 (29+ documents, 9 analysis dates, 50+ biological parameters), the tool identified critical findings including an unreported drug interaction (Haloperidol + Donepezil double QTc prolongation), chronic untreated vitamin D deficiency, complete left internal carotid artery occlusion, and insufficient renal monitoring under Dabigatran. Implemented as a single-file PWA in pure HTML5/JavaScript, running 100% client-side. GNU GPL v3.

Keywords: longitudinal medical record, pharmacovigilance, OCR, PWA, patient empowerment, clinical decision support, open source, LLM-assisted extraction

1. Introduction

The fragmentation of medical records across multiple healthcare providers remains a critical barrier to effective longitudinal patient care. Patients with complex, multi-system pathologies frequently accumulate documents from various hospitals, specialists and laboratories, with no single tool providing a unified chronological view. MediTrace addresses this gap with a radically simple architecture: a single HTML file, running entirely in the browser, requiring no server, no account, no installation.

2. New Features in v2026.3

2.1 Client-Side OCR (Tesseract.js)

MediTrace integrates Tesseract.js v4 for client-side optical character recognition, supporting Spanish as primary language with English fallback. For scanned PDFs, pdf.js (Mozilla) renders each page to a high-resolution canvas (2x scale) before OCR processing. A dedicated "OCR-only" mode allows text extraction without an API key.

2.2 Claude.ai Workflow Without API Key

A streamlined workflow for users with a Claude.ai subscription: after OCR extraction, "Copy for Claude" prepares the OCR text with the extraction prompt. The user pastes into claude.ai, copies the JSON, and uses "Paste and import from clipboard" for one-click integration.

2.3 Bacteriology Module

A new panel tracks culture results and antibiograms over time: specimen type, identified organism, colony count, and structured antibiogram with S/I/R classification per antibiotic.

2.4 Radar Chart

A normalised overview of all biological parameters. Each axis represents a parameter normalised to its reference range (100% = middle of normal). Multiple dates can be superimposed. Eight predefined parameter groups allow focused analysis.

2.5 Progressive Web App (PWA)

MediTrace can be installed as a standalone application on mobile and desktop. A service worker caches all assets for offline use (network-first strategy with cache fallback).

2.6 Extensible Parameters and Source Memory

Users can create custom biological parameters and previously used sources are memorised for rapid re-selection.

2.7 Accessibility: Light Theme and Enlarged Text

A high-contrast light theme (cream background, dark text, bold labels) with increased font sizes (18px base). Driven by real user feedback from the primary caregiver (age 83).

3. Technical Architecture

Component	Technology	Purpose
Frontend	HTML5/CSS3/ES6+	Single-file application
Charts	Chart.js 4.4	Line charts, radar charts
OCR	Tesseract.js 4.x	Client-side text recognition
PDF rendering	pdf.js 3.11	Scanned PDF to canvas
AI extraction	Claude Sonnet 4.6	Structured JSON from text
Storage	localStorage	Measures, events, meds, bacterio
Offline	Service Worker	Network-first + cache fallback
Install	PWA manifest	Standalone mobile/desktop app
Backup	.mt (JSON)	Complete data export/import

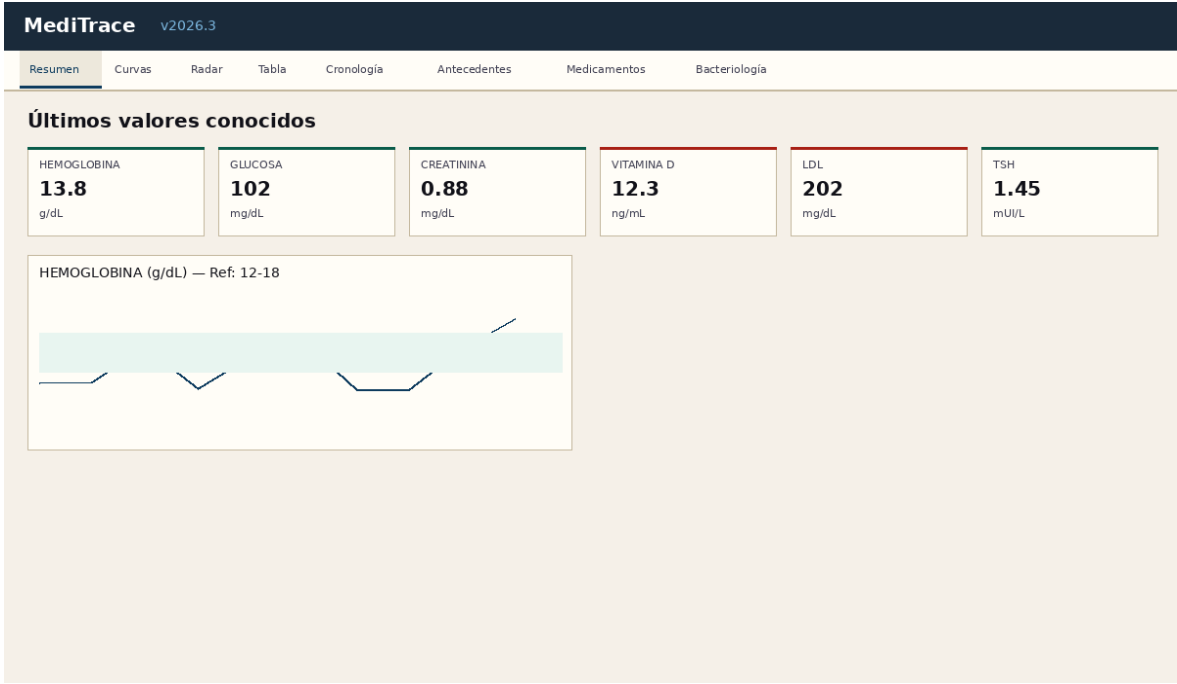


Figure 1: Summary view — KPI cards with latest biological values and evolution chart.

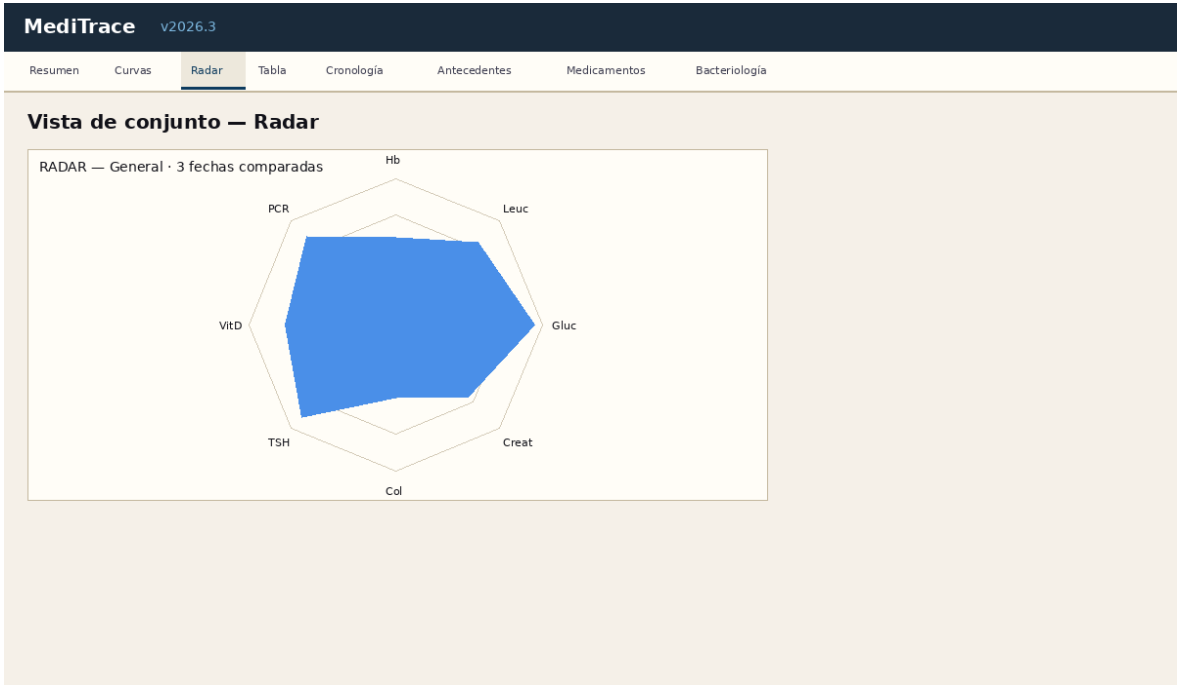


Figure 2: Radar chart — Normalised parameter overview with multiple dates superimposed.

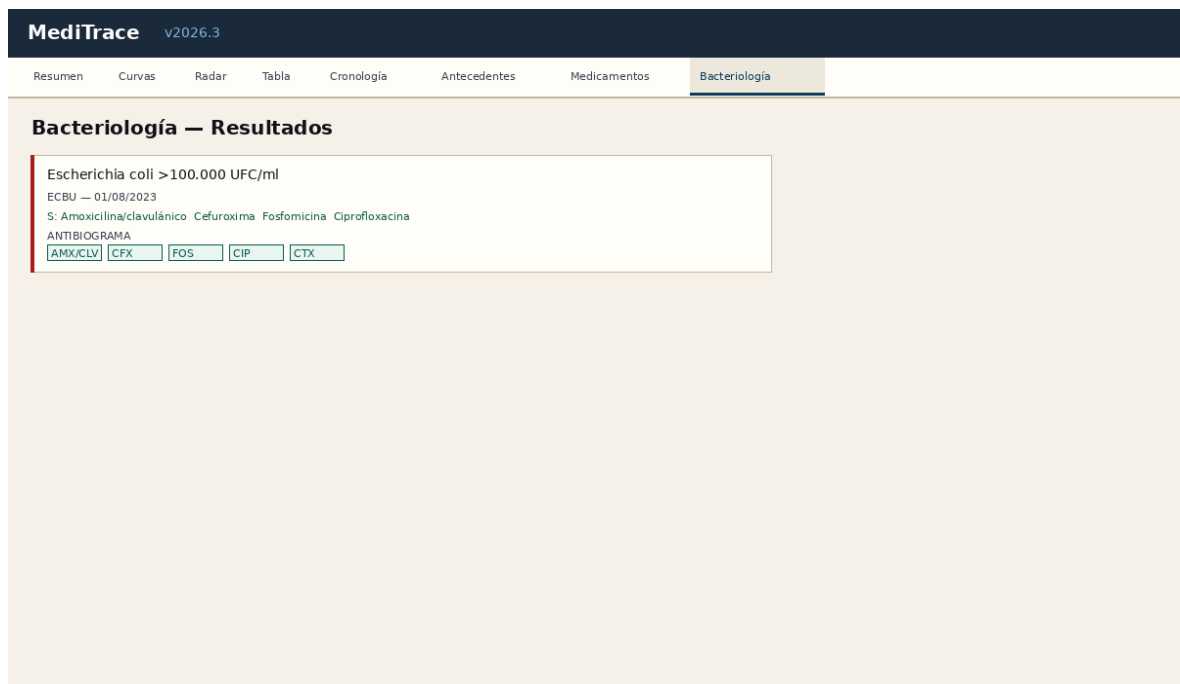


Figure 3: Bacteriology module — Culture result with structured antibiogram (S/I/R).

4. Clinical Case

Applied to a real patient (F, 87y) with multi-system pathology spanning 2016-2026: 29+ documents, 9 analysis dates, 50+ biological parameters, 12+ medications, 15+ clinical events.

4.1 Key Clinical Findings

- **Complete left ICA occlusion** (MRI March 2026) — retrospectively correlated with left frontal hypometabolism on PET/CT from December 2021.
- **QTc prolongation risk** — Haloperidol + Donepezil co-prescription identified. Haloperidol subsequently discontinued.
- **Anticoagulation abandonment** — Dabigatran discontinued by family (Feb 2025) with recurrent PE and carotid occlusion.
- **Chronic vitamin D deficiency** — 27.10 to 12.33 ng/mL. Hydroferol supplementation initiated.
- **Cognitive decline** — MMSE: 25 to 16 (2021-2024). PET/CT pattern: vascular/frontotemporal, not Alzheimer's.
- **Renal function decline** — GFR: 83 to 70.8 mL/min. Mandatory monitoring under Dabigatran.

5. Discussion

MediTrace embodies the principle: "Why do what the computer can do?" Every manual step has been systematically eliminated. The tool is designed for elderly caregivers with limited technical expertise. Unlike cloud-based EHR systems, MediTrace processes all data locally. The entire application was co-developed through iterative conversation between the caregiver and Claude (LLM), demonstrating that meaningful clinical tools can be built without traditional software development teams.

6. Future Work

- Batch OCR processing
- Global search across all data
- Print optimisation with large fonts
- Encrypted .mt backup files

- Enriched bacteriology with resistance evolution
- Active alert badges

7. Version History

Version	Date	Changes
v2026.1	May 2026	Initial release. PDF extraction, charts, pharmacovigilance.
v2026.2	May 2026	Medications, observance, correlations, batch PDF, glossary.
v2026.3	May 2026	OCR, bacteriology, radar, PWA, light theme, extensible params, landing page, guide.

8. References

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